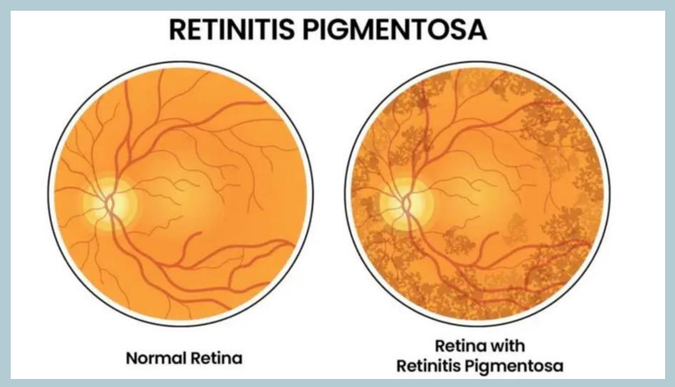
Comparative Retinal Transcriptomics Across Model Organisms to Inform Human Retinitis Pigmentosa Research

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Introduction

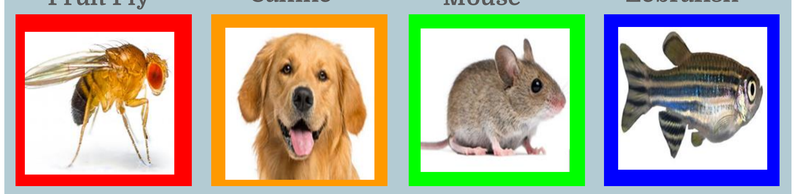
Retinitis Pigmentosa (RP) is a genetically inherited retinal disorder marked by progressive photoreceptor degeneration, leading to night blindness, tunnel vision, and eventual vision loss. Although it affects more than 2 million people worldwide, there is currently no cure. RP is studied across a wide range of model organisms, each offering unique insights into retinal biology and disease mechanisms.



Here, we compare four species, *zebrafish*, *fruit fly*, *canine*, and *mouse*, to evaluate their utility in modeling RP in humans. We integrate genome-wide association studies (GWAS) with species-specific retinal RNA-seq data to identify conserved genes and pathways implicated in RP pathogenesis. By performing cross-species differential expression analysis (DEA), gene ontology (GO) enrichment, and ortholog mapping, we highlight both shared and divergent molecular mechanisms.

Our goal is to rank and contextualize each model organism's translational relevance in RP research, ultimately guiding future experimental design and therapeutic development.

Fruit Fly | Canine | Mouse | Zebrafish



Methods

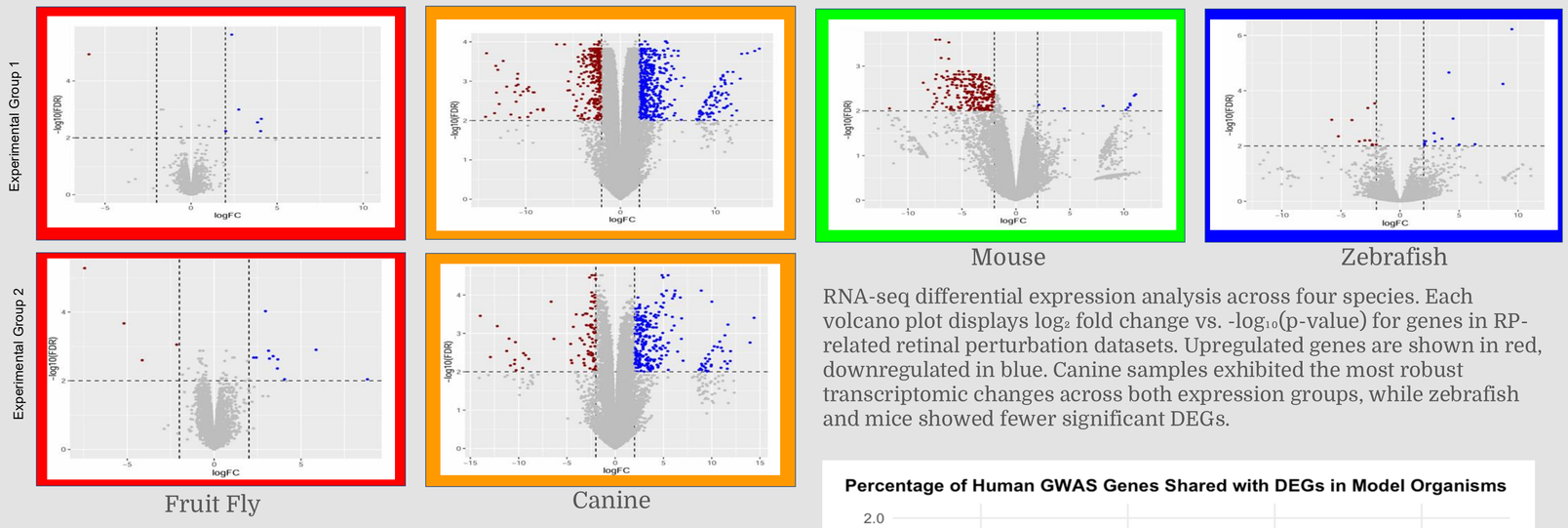
1) **RNA-seq Differential Expression Analysis:**
- Retrieved retinal RNA-seq datasets for *human,* (GSE213807) *mouse* (GSE56473), *zebrafish* (GSE101544), *dog* (GSE97638), and *fly* (GSE139361) from GEO/SRA.
- Performed quality control analysis using FastQC, trimmed low-quality reads using TrimGalore.
- Quantified transcript abundance using Salmon
- Used DESeq2 for differential expression analysis, comparing RP vs. Control Samples.

1) **GWAS Annotation & Gene Mapping:**
- Filtered annotated RP-related genes from the NHGRI-EBI GWAS Catalog ($p < 5 \times 10^{-8}$)
- Use biomaRt R package to map GWAS genes to 1-1 orthologs in each species, and compared with genes identified from DEA.

1) **Pathway Enrichment Analysis:**
- Use clusterProfiler enrichment on curated gene sets identified from GWAS and DEA to identify conserved RP-relevant pathways (ex. phototransduction, ciliary transport, etc.).

Results

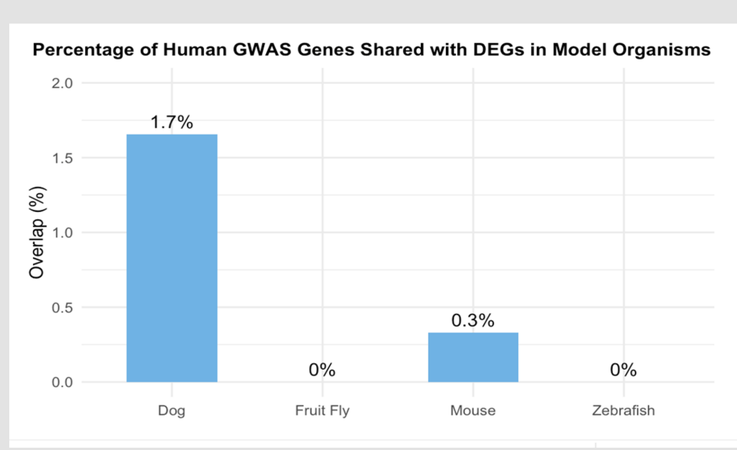
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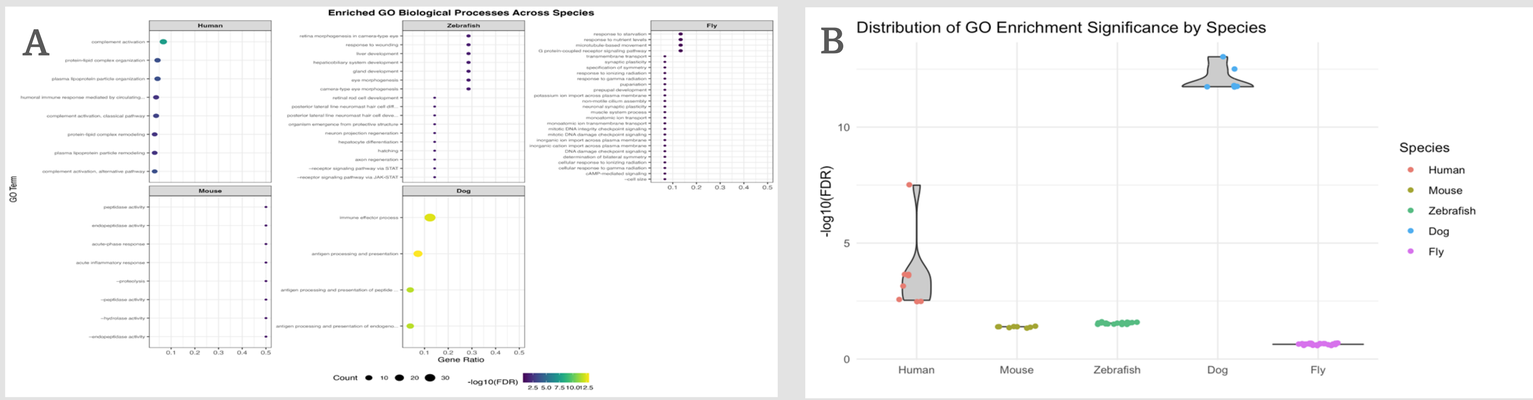
RNA-seq differential expression analysis across four species. Each volcano plot displays $\log_2$ fold change vs. $-\log_{10}$(p-value) for genes in RP-related retinal perturbation datasets. Upregulated genes are shown in red, downregulated in blue. Canine samples exhibited the most robust transcriptomic changes across both expression groups, while zebrafish and mice showed fewer significant DEGs.

2) **GWAS Annotation & Gene Mapping:**

Overlap of human genome-wide significant RP-associated genes (n=302) with differentially expressed genes in model organism retinas. Dog had 5 overlapping genes (1.66%), mouse had 1 overlapping gene (0.33%) while zebrafish and drosophila showed no overlap.



3) **Pathway Enrichment Analysis:**



GO enrichment analysis of differentially expressed genes across five species reveals distinct biological themes and varying significance. Figure A shows the top enriched GO terms per species. Dog genes were strongly enriched for antigen processing pathways with -log10(FDR) > 12.5, while other species showed weaker enrichment for lipid transport (human), inflammation (mouse), development (zebrafish), and stress response (fly). Figure B highlights the higher enrichment significance in dog compared to all other species.

Discussion & Conclusions

- RNA-seq analysis reveals species-specific transcriptional responses to RP, with Canine and Mouse showing the most robust up & down-regulated gene sets.
- Overlap with human GWAS was minimal, with canines showing the most overlap at only 1.66%.
- GO pathway analysis shows strong enrichment in canines and a unique role of antigen processing in retinal degeneration, compared to weaker and more variable enrichment patterns in other species.
- Overall, canines demonstrated the strongest transcriptional signal with a high number of DEGs and most significant GO enrichment, suggesting that it may be the best model to recapitulate RP-related retinal pathology.

Limitations & Implications

- Some RNA-seq datasets had limited biological replicates, such as Zebrafish, reducing statistical power and leading to false negatives or underestimation of differential expression.
- Practical considerations like cost, availability, and ethical constraints vary across model organisms, as dogs showed the strongest transcriptomic and GO signal but are less accessible than mice or flies, which may affect their feasibility for large-scale or high-throughput RP studies.

References:

https://docs.google.com/document/d/1DdoZC8DE45iTuYmun_idigHtWFevdmh2XhpjLdPVWLg/edit?usp=sharing

Acknowledgements

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